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ECE 2140 – 30

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# Final Project Report

## 1. Introduction

The purpose of this final project is to be able to implement the knowledge we learned in lectures such as truth tables, circuit implementation, K-Maps, and more, along with everything we learned in the previous labs and apply them to our final project designs. This final project required implementing a breadboard aspect along with the use of Verilog code to program the Basys device.

## 2. Specifications

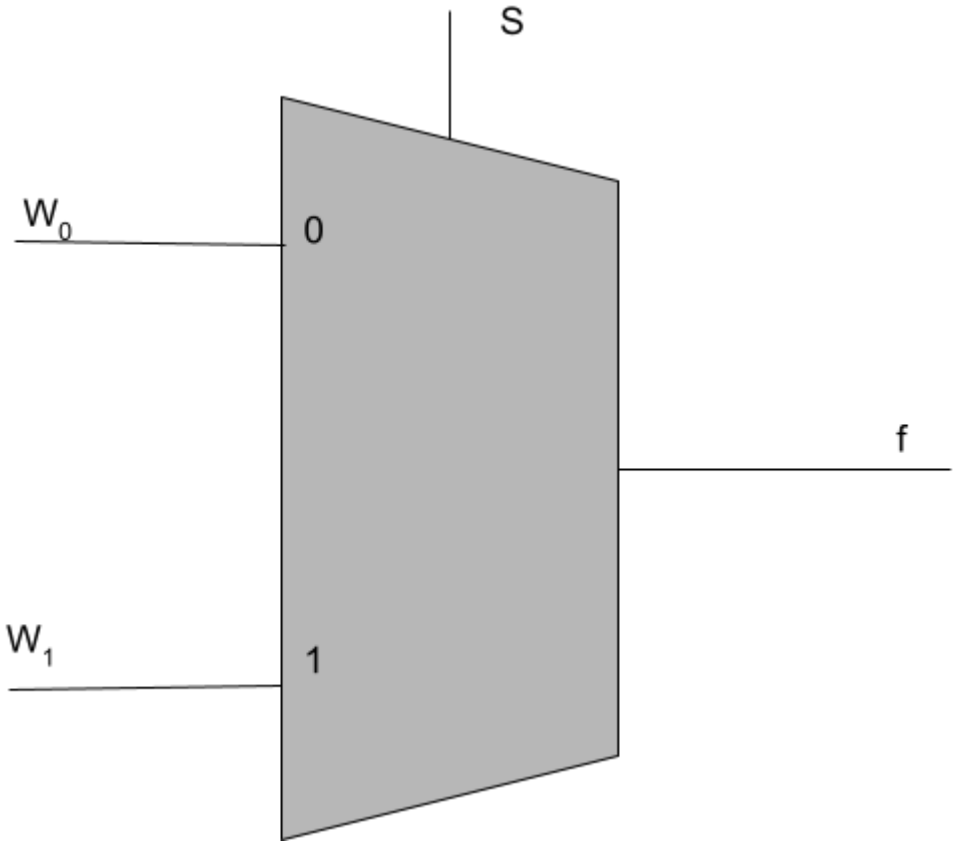
For this final project I was tasked with two separate implementations. The first involved a breadboard implementation that focused on building a 2-to-1 MUX by using any needed gates to implement the circuit on a breadboard and then display the inputs and outputs using LEDs. The second implementation involved using Verilog code in which I chose my own idea which was approved by the TA. The description of the functionality of this idea can be found in section **4.2** along with the code.

3. **Design**

**2-1 MUX:**

**Inputs:** $W_0$,$W_1$, S

**Outputs:** f



**Truth Tables:**

| S | f |
|---|---|
| 0 | $W_0$ |
| 1 | $W_1$ |

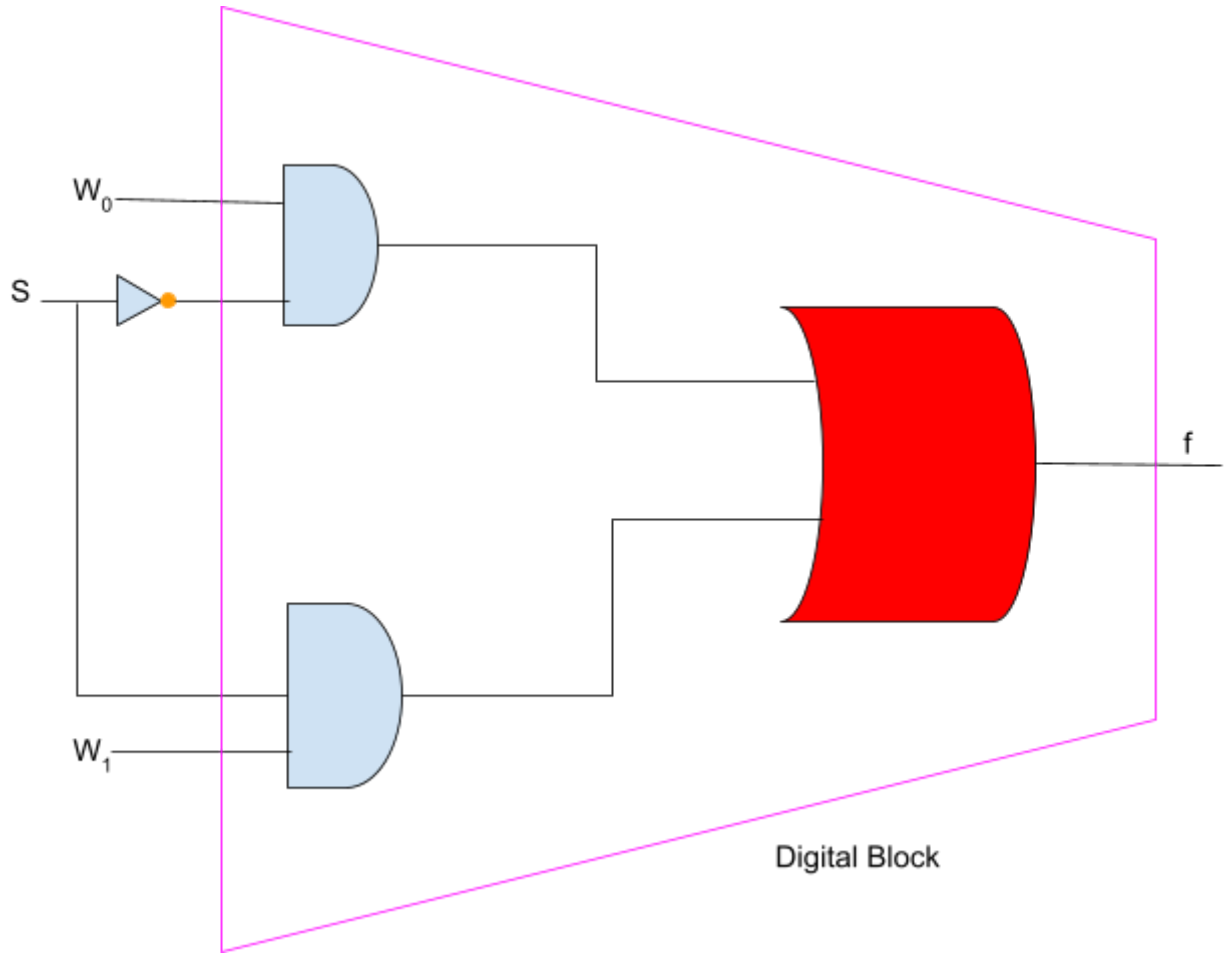
| S | $W_0$ | $W_1$ | f |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

**K-Map:**

| S \ $W_0W_1$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 |

$f(S,W_0,W_1) = SW_1 + \sim SW_0$

**Circuit Design:**



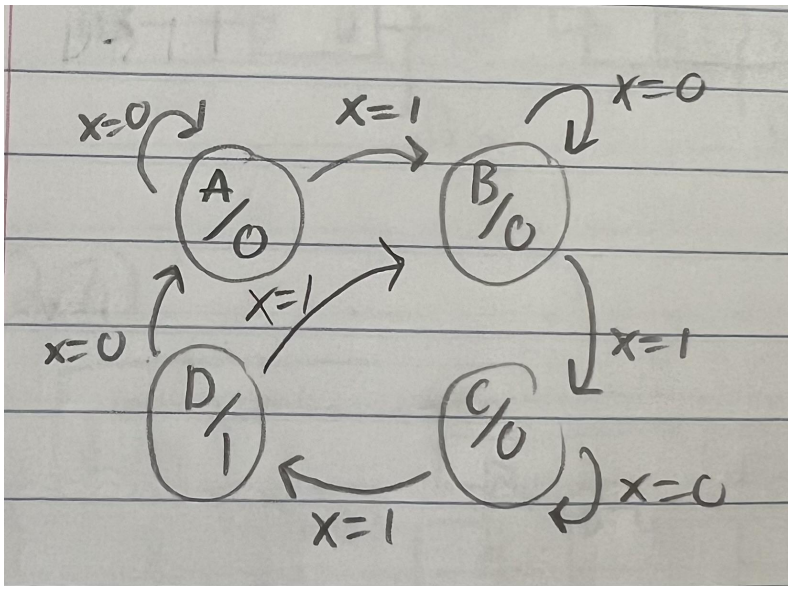
**Gates:**

- 2 AND Gates
- 1 OR Gate
- 1 Inverter

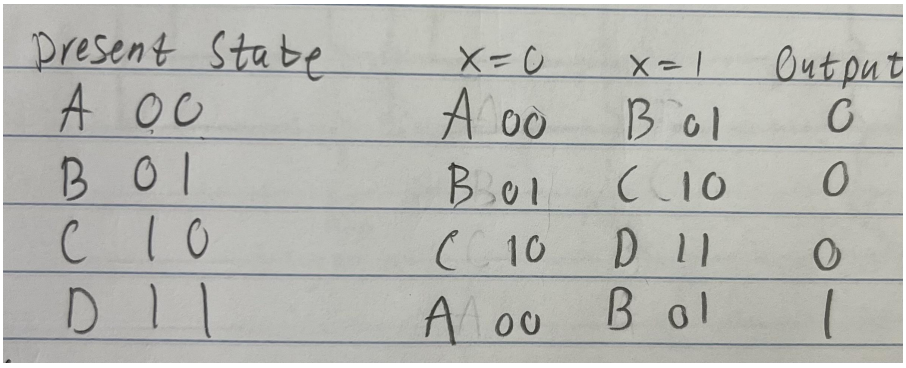
## 4. Implementation

### 4.1-2 - Verilog project:

**Design of the vending FSM:**



**State Table:**



| Present State | x=0 | x=1 | Output |
|---|---|---|---|
| A 00 | A 00 | B 01 | 0 |
| B 01 | B 01 | C 10 | 0 |
| C 10 | C 10 | D 11 | 0 |
| D 11 | A 00 | B 01 | 1 |

The purpose of this part of the project is similar to lab 10, there are 4 States, A-D, that display the numbers 1,2,3,4 for each anode. The twist is that when the user holds the dime button to go to the next state, all the numbers flash at the same time, and when the user lets go the next state will show.

To explain how the code below works, this is very similar to lab 10 except with a few adjustments. The difference in this code is that the anodes are declared as “output reg” because this allows me to have control of the anodes that are dependent on the current state. The most efficient and organized way I implemented this is by placing anode0-3 into an array, and I would turn on the corresponding anode by using bits, similar to how we placed a-g in an array to make our numbers using the LEDs. I also made the clock faster which allows the user when he holds the input dime button, to look like all the numbers are flashing at the same time.

**Verilog Code:**

```
`timescale 1ns / 1ps

module Verilog_Final4(dime,clk,a,b,c,d,e,f,g,reset,tealed,anode0,anode1,anode2,anode3);
input dime,clk, reset;
output reg tealed,a,b,c,d,e,f,g;
output reg anode0,anode1,anode2,anode3;
```

```
reg [1:0] Statecurrent;
reg [1:0] Statenext;
reg [27:0] counter;
reg enable;

parameter Astate = 2'b00, Bstate = 2'b01, Cstate = 2'b10, Dstate = 2'b11;

always@(Statecurrent, dime)
   case(Statecurrent)
      Astate: if (dime ==1) Statenext <= Bstate;
            else Statenext <= Astate;
      Bstate: if (dime ==1) Statenext <= Cstate;
            else Statenext <= Bstate;
      Cstate: if (dime ==1) Statenext <= Dstate;
            else Statenext <= Cstate;
      Dstate: if (dime ==1) Statenext <= Astate;
            else Statenext <= Dstate;

endcase

always@(Statecurrent)
case(Statecurrent)

   Astate: tealed <=0;
   Bstate: tealed <=0;
   Cstate: tealed <=0;
   Dstate: tealed <=1;

endcase



always@(Statecurrent)
begin
   case(Statecurrent)
       Astate: {a,b,c,d,e,f,g} = 'b1001111;
       Bstate: {a,b,c,d,e,f,g} = 'b0010010;
       Cstate: {a,b,c,d,e,f,g} = 'b0000110;
       Dstate: {a,b,c,d,e,f,g} = 'b1001100;

 endcase
```

```
 end

 always @ (posedge clk)
begin
if (counter == 28'd0) counter <= 28'd01_000_000;
else counter <= counter - 1'd1;
enable <= counter == 28'd0;
end

always @ (posedge clk)
begin
if (reset==1)
Statecurrent<=Astate; //update of the state if reset
else
if (enable ==1)
Statecurrent <= Statenext; //update of the state without reset
end


always@(Statecurrent)
begin
    case(Statecurrent)
    Astate: {anode0,anode1,anode2,anode3} = 'b1110;
    Bstate: {anode0,anode1,anode2,anode3} = 'b1101;
    Cstate: {anode0,anode1,anode2,anode3} = 'b1011;
    Dstate: {anode0,anode1,anode2,anode3} = 'b0111;

endcase
end

endmodule
```

## **XDC File:**

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal
names in the project

## Clock signal
set_property PACKAGE_PIN W5 [get_ports clk]
        set_property IOSTANDARD LVCMOS33 [get_ports clk]
```

```
        create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]

## Switches
#set_property PACKAGE_PIN V17 [get_ports {sw[0]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[0]}]
#set_property PACKAGE_PIN V16 [get_ports {sw[1]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[1]}]
#set_property PACKAGE_PIN W16 [get_ports {sw[2]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[2]}]
#set_property PACKAGE_PIN W17 [get_ports {sw[3]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[3]}]
#set_property PACKAGE_PIN W15 [get_ports {sw[4]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[4]}]
#set_property PACKAGE_PIN V15 [get_ports {sw[5]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[5]}]
#set_property PACKAGE_PIN W14 [get_ports {sw[6]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[6]}]
#set_property PACKAGE_PIN W13 [get_ports {sw[7]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[7]}]
#set_property PACKAGE_PIN V2 [get_ports {sw[8]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[8]}]
#set_property PACKAGE_PIN T3 [get_ports {sw[9]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[9]}]
#set_property PACKAGE_PIN T2 [get_ports {sw[10]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[10]}]
#set_property PACKAGE_PIN R3 [get_ports {sw[11]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[11]}]
#set_property PACKAGE_PIN W2 [get_ports {sw[12]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[12]}]
#set_property PACKAGE_PIN U1 [get_ports {sw[13]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[13]}]
#set_property PACKAGE_PIN T1 [get_ports {sw[14]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[14]}]
#set_property PACKAGE_PIN R2 [get_ports {sw[15]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {sw[15]}]


## LEDs
set_property PACKAGE_PIN U16 [get_ports {tealed}]
        set_property IOSTANDARD LVCMOS33 [get_ports {tealed}]
#set_property PACKAGE_PIN E19 [get_ports {led[1]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[1]}]
#set_property PACKAGE_PIN U19 [get_ports {led[2]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[2]}]
```

```
#set_property PACKAGE_PIN V19 [get_ports {led[3]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[3]}]
#set_property PACKAGE_PIN W18 [get_ports {led[4]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[4]}]
#set_property PACKAGE_PIN U15 [get_ports {led[5]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[5]}]
#set_property PACKAGE_PIN U14 [get_ports {led[6]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[6]}]
#set_property PACKAGE_PIN V14 [get_ports {led[7]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[7]}]
#set_property PACKAGE_PIN V13 [get_ports {led[8]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[8]}]
#set_property PACKAGE_PIN V3 [get_ports {led[9]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[9]}]
#set_property PACKAGE_PIN W3 [get_ports {led[10]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[10]}]
#set_property PACKAGE_PIN U3 [get_ports {led[11]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[11]}]
#set_property PACKAGE_PIN P3 [get_ports {led[12]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[12]}]
#set_property PACKAGE_PIN N3 [get_ports {led[13]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[13]}]
#set_property PACKAGE_PIN P1 [get_ports {led[14]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[14]}]
#set_property PACKAGE_PIN L1 [get_ports {led[15]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {led[15]}]


##7 segment display
set_property PACKAGE_PIN W7 [get_ports {a}]
        set_property IOSTANDARD LVCMOS33 [get_ports {a}]
set_property PACKAGE_PIN W6 [get_ports {b}]
        set_property IOSTANDARD LVCMOS33 [get_ports {b}]
set_property PACKAGE_PIN U8 [get_ports {c}]
        set_property IOSTANDARD LVCMOS33 [get_ports {c}]
set_property PACKAGE_PIN V8 [get_ports {d}]
        set_property IOSTANDARD LVCMOS33 [get_ports {d}]
set_property PACKAGE_PIN U5 [get_ports {e}]
        set_property IOSTANDARD LVCMOS33 [get_ports {e}]
set_property PACKAGE_PIN V5 [get_ports {f}]
        set_property IOSTANDARD LVCMOS33 [get_ports {f}]
set_property PACKAGE_PIN U7 [get_ports {g}]
        set_property IOSTANDARD LVCMOS33 [get_ports {g}]
```

```
#set_property PACKAGE_PIN V7 [get_ports dp]
        #set_property IOSTANDARD LVCMOS33 [get_ports dp]

set_property PACKAGE_PIN U2 [get_ports {anode0}]
        set_property IOSTANDARD LVCMOS33 [get_ports {anode0}]
set_property PACKAGE_PIN U4 [get_ports {anode1}]
        set_property IOSTANDARD LVCMOS33 [get_ports {anode1}]
set_property PACKAGE_PIN V4 [get_ports {anode2}]
        set_property IOSTANDARD LVCMOS33 [get_ports {anode2}]
set_property PACKAGE_PIN W4 [get_ports {anode3}]
        set_property IOSTANDARD LVCMOS33 [get_ports {anode3}]


##Buttons
set_property PACKAGE_PIN U18 [get_ports reset]
        set_property IOSTANDARD LVCMOS33 [get_ports reset]
#set_property PACKAGE_PIN T18 [get_ports btnU]
        #set_property IOSTANDARD LVCMOS33 [get_ports btnU]
#set_property PACKAGE_PIN W19 [get_ports btnL]
        #set_property IOSTANDARD LVCMOS33 [get_ports btnL]
set_property PACKAGE_PIN T17 [get_ports dime]
        set_property IOSTANDARD LVCMOS33 [get_ports dime]
#set_property PACKAGE_PIN U17 [get_ports btnD]
        #set_property IOSTANDARD LVCMOS33 [get_ports btnD]



##Pmod Header JA
##Sch name = JA1
#set_property PACKAGE_PIN J1 [get_ports {JA[0]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JA[0]}]
##Sch name = JA2
#set_property PACKAGE_PIN L2 [get_ports {JA[1]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JA[1]}]
##Sch name = JA3
#set_property PACKAGE_PIN J2 [get_ports {JA[2]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JA[2]}]
##Sch name = JA4
#set_property PACKAGE_PIN G2 [get_ports {JA[3]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JA[3]}]
##Sch name = JA7
#set_property PACKAGE_PIN H1 [get_ports {JA[4]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JA[4]}]
##Sch name = JA8
```

```
#set_property PACKAGE_PIN K2 [get_ports {JA[5]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JA[5]}]
##Sch name = JA9
#set_property PACKAGE_PIN H2 [get_ports {JA[6]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JA[6]}]
##Sch name = JA10
#set_property PACKAGE_PIN G3 [get_ports {JA[7]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JA[7]}]



##Pmod Header JB
##Sch name = JB1
#set_property PACKAGE_PIN A14 [get_ports {JB[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[0]}]
##Sch name = JB2
#set_property PACKAGE_PIN A16 [get_ports {JB[1]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[1]}]
##Sch name = JB3
#set_property PACKAGE_PIN B15 [get_ports {JB[2]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[2]}]
##Sch name = JB4
#set_property PACKAGE_PIN B16 [get_ports {JB[3]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[3]}]
##Sch name = JB7
#set_property PACKAGE_PIN A15 [get_ports {JB[4]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[4]}]
##Sch name = JB8
#set_property PACKAGE_PIN A17 [get_ports {JB[5]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[5]}]
##Sch name = JB9
#set_property PACKAGE_PIN C15 [get_ports {JB[6]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[6]}]
##Sch name = JB10
#set_property PACKAGE_PIN C16 [get_ports {JB[7]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JB[7]}]



##Pmod Header JC
##Sch name = JC1
#set_property PACKAGE_PIN K17 [get_ports {JC[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JC[0]}]
##Sch name = JC2
```

```
#set_property PACKAGE_PIN M18 [get_ports {JC[1]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[1]}]
##Sch name = JC3
#set_property PACKAGE_PIN N17 [get_ports {JC[2]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[2]}]
##Sch name = JC4
#set_property PACKAGE_PIN P18 [get_ports {JC[3]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[3]}]
##Sch name = JC7
#set_property PACKAGE_PIN L17 [get_ports {JC[4]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[4]}]
##Sch name = JC8
#set_property PACKAGE_PIN M19 [get_ports {JC[5]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[5]}]
##Sch name = JC9
#set_property PACKAGE_PIN P17 [get_ports {JC[6]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[6]}]
##Sch name = JC10
#set_property PACKAGE_PIN R18 [get_ports {JC[7]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JC[7]}]


##Pmod Header JXADC
##Sch name = XA1_P
#set_property PACKAGE_PIN J3 [get_ports {JXADC[0]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[0]}]
##Sch name = XA2_P
#set_property PACKAGE_PIN L3 [get_ports {JXADC[1]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[1]}]
##Sch name = XA3_P
#set_property PACKAGE_PIN M2 [get_ports {JXADC[2]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[2]}]
##Sch name = XA4_P
#set_property PACKAGE_PIN N2 [get_ports {JXADC[3]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[3]}]
##Sch name = XA1_N
#set_property PACKAGE_PIN K3 [get_ports {JXADC[4]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[4]}]
##Sch name = XA2_N
#set_property PACKAGE_PIN M3 [get_ports {JXADC[5]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[5]}]
##Sch name = XA3_N
#set_property PACKAGE_PIN M1 [get_ports {JXADC[6]}]
        #set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[6]}]
```

```
##Sch name = XA4_N
#set_property PACKAGE_PIN N1 [get_ports {JXADC[7]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {JXADC[7]}]



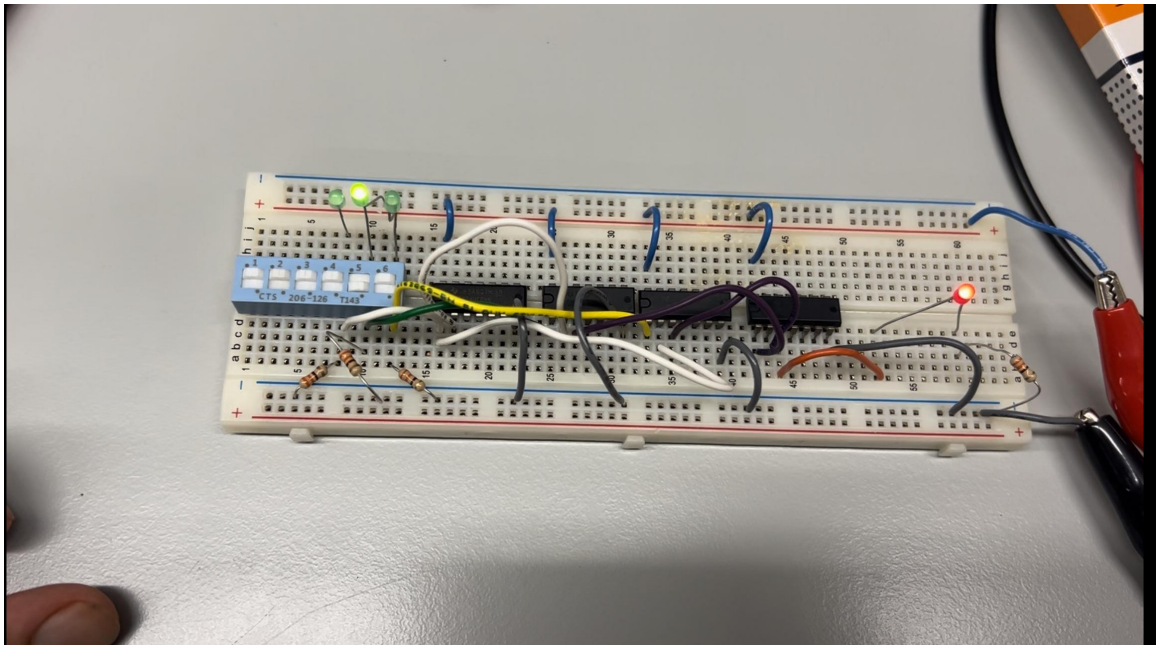
##VGA Connector
#set_property PACKAGE_PIN G19 [get_ports {vgaRed[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaRed[0]}]
#set_property PACKAGE_PIN H19 [get_ports {vgaRed[1]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaRed[1]}]
#set_property PACKAGE_PIN J19 [get_ports {vgaRed[2]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaRed[2]}]
#set_property PACKAGE_PIN N19 [get_ports {vgaRed[3]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaRed[3]}]
#set_property PACKAGE_PIN N18 [get_ports {vgaBlue[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaBlue[0]}]
#set_property PACKAGE_PIN L18 [get_ports {vgaBlue[1]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaBlue[1]}]
#set_property PACKAGE_PIN K18 [get_ports {vgaBlue[2]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaBlue[2]}]
#set_property PACKAGE_PIN J18 [get_ports {vgaBlue[3]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaBlue[3]}]
#set_property PACKAGE_PIN J17 [get_ports {vgaGreen[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaGreen[0]}]
#set_property PACKAGE_PIN H17 [get_ports {vgaGreen[1]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaGreen[1]}]
#set_property PACKAGE_PIN G17 [get_ports {vgaGreen[2]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaGreen[2]}]
#set_property PACKAGE_PIN D17 [get_ports {vgaGreen[3]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {vgaGreen[3]}]
#set_property PACKAGE_PIN P19 [get_ports Hsync]
	#set_property IOSTANDARD LVCMOS33 [get_ports Hsync]
#set_property PACKAGE_PIN R19 [get_ports Vsync]
	#set_property IOSTANDARD LVCMOS33 [get_ports Vsync]


##USB-RS232 Interface
#set_property PACKAGE_PIN B18 [get_ports RsRx]
	#set_property IOSTANDARD LVCMOS33 [get_ports RsRx]
#set_property PACKAGE_PIN A18 [get_ports RsTx]
	#set_property IOSTANDARD LVCMOS33 [get_ports RsTx]
```

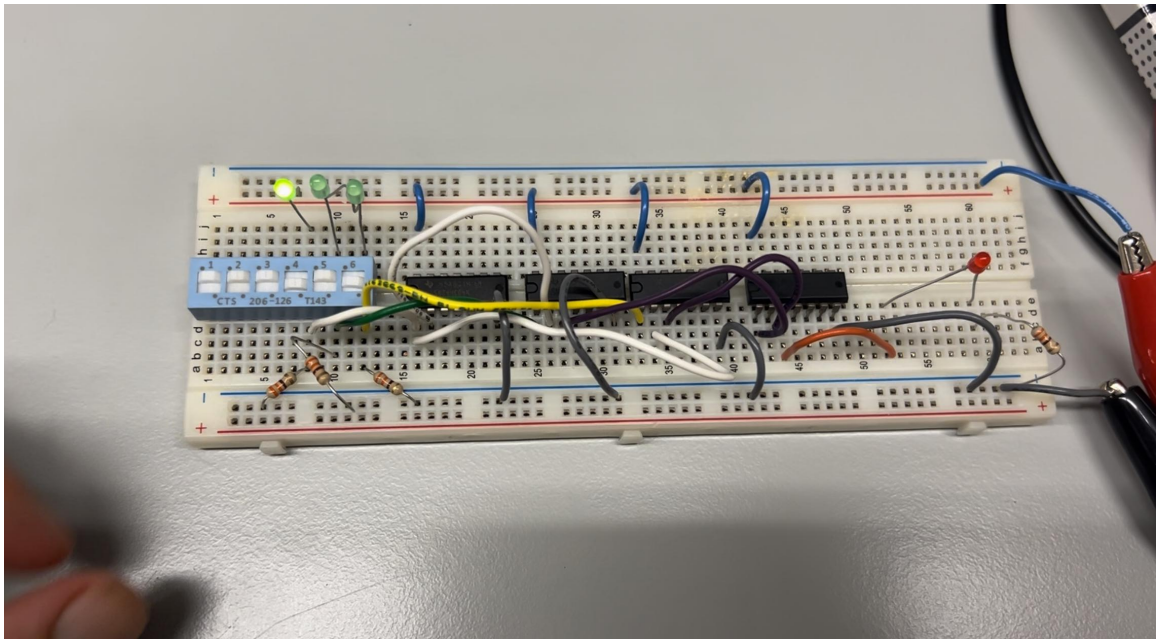
##USB HID (PS/2)
#set_property PACKAGE_PIN C17 [get_ports PS2Clk]
	#set_property IOSTANDARD LVCMOS33 [get_ports PS2Clk]
	#set_property PULLUP true [get_ports PS2Clk]
#set_property PACKAGE_PIN B17 [get_ports PS2Data]
	#set_property IOSTANDARD LVCMOS33 [get_ports PS2Data]
	#set_property PULLUP true [get_ports PS2Data]

##Quad SPI Flash
##Note that CCLK_0 cannot be placed in 7 series devices. You can access it using the
##STARTUPE2 primitive.
#set_property PACKAGE_PIN D18 [get_ports {QspiDB[0]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {QspiDB[0]}]
#set_property PACKAGE_PIN D19 [get_ports {QspiDB[1]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {QspiDB[1]}]
#set_property PACKAGE_PIN G18 [get_ports {QspiDB[2]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {QspiDB[2]}]
#set_property PACKAGE_PIN F18 [get_ports {QspiDB[3]}]
	#set_property IOSTANDARD LVCMOS33 [get_ports {QspiDB[3]}]
#set_property PACKAGE_PIN K19 [get_ports QspiCSn]
	#set_property IOSTANDARD LVCMOS33 [get_ports QspiCSn]

## 4.3 - Breadboard Implementation:



The first gate is the inverter, followed by 2 AND Gates and at the very end is an OR Gate. This is supposed to follow the circuit design shown previously. The switch at the very left of the breadboard allows the user to turn on each input, and the green LEDs are used as indicators of them being on. The red LED represents whether the output is on or off. The green LEDs are in order with s being the first input, followed by $W_0$, then $W_1$. In this instance shown above, $W_0$ is turned on, so the output is on which supports the truth table shown previously. The instance for the picture below shows $W_1$ being on, however the output is off, once again following the truth table.



## 5. Conclusion:

This final project required the implementation of all the course material in order to make both the breadboard and Verilog portions of the project to work. After completing these projects, in the future, I would look into working on using even more states to maybe create phrases or sentences by applying letters to each state, and then putting the states into an array like how I did for the anodes, to create words using a combination of states. Then I would probably declare each word into a state to make a sentence after the user pushes one of the input buttons to go to the next state. In conclusion, this final project allowed us to see our concept of the material work before our eyes and find different ways to create other logic systems in the future.